

PICO Framework: The Optimal Firm Size for SME DT Training — An Inverted-U Effect

Date: 2026-05-01
Project: SME Transformation (K-Hi-Tech Platform, 4-year program)
Strategy: Strategy 3 — Firm-Size Inverted U and Tailored Training Strategy (Highest Practical Value)

Research Question

Does the effectiveness of SME digital transformation (DT) training have a nonlinear (inverted-U) relationship with firm size, and does an optimal size point exist?

Strategy 3: Firm-Size Inverted U and Tailored Training Strategy (Highest Practical Value)

Item	Content
Topic	The optimal firm size for SME DT training: inverted-U effect and differentiated support strategy
Core Analyses	M5 (moderated regression) + T15 (ANOVA) + T16 (moderated mediation) + B1/C1 (RSM inverted U)
Theory	Liability of Smallness + Resource Orchestration + Optimal Distinctiveness
Variables	IV: Firm size × DT awareness / DV: Q3_Diff
Key Findings	Firm-size inverted U $p = .029$, optimum ≈ 13 employees. Large-firm paradox. SF + multiple participation shows negative synergy
Theoretical Contribution	Nonlinear extension of LoS. Empirical evidence of an Optimal Level in training effectiveness
Methodological Contribution	Analytical deepening strategy from OLS moderation to RSM inverted U
Strengths	Clear policy implications (tailored support), practically applicable, novel inverted-U finding
Weaknesses	Low R^2 , crude firm-size proxy (number of regular employees)

Target Journals:

Rank	Journal	Level	Rationale
1	<i>Small Business Economics</i>	SSCI Q1 (IF ~7)	Core journal for SMEs and size effects
2	<i>International Small Business Journal</i>	SSCI Q1 (IF ~7)	SME training / innovation
3	<i>Technovation</i>	SSCI Q1 (IF ~12)	Technological innovation + SMEs + policy
4	<i>Journal of Small Business Management</i>	SSCI Q1 (IF ~7)	SME management + HRD
5	<i>R&D Management</i>	SSCI Q1 (IF ~5)	R&D investment optimization

P — Population

Category	Content
Population	SMEs participating in the K-Hi-Tech Platform (n = 282)
Segmentation	By number of regular employees — micro enterprises (<5), small enterprises (5–50), medium enterprises (50–300)
Inclusion Criteria	Participated in DT training + Q3_Diff measurable + firm-size data available
Stratification Variables	Industry group, sales volume, DT maturity stage
Key Observation	Estimated optimal size ≈ 13 employees

I — Intervention

Category	Content
Primary Treatment	DT training participation × DT awareness level (interaction)
Moderator	Firm size (number of regular employees, linear + quadratic term)
Additional Treatment	SF (Smart Factory track) + multiple participation combination
Theoretical Basis	Liability of Smallness + Resource Orchestration + Optimal Distinctiveness — an optimum between resource scarcity (small firms) and inertia (large firms)

C — Comparator

Category	Content
Primary Comparison	Training effectiveness in small vs. medium vs. large firms
Secondary Comparison	Firms around the optimal size (≈13 employees) vs. all other firms
Tertiary Comparison	SF-only participation vs. SF + multiple participation (testing negative synergy)
Analytical Strategy	Stepwise deepening: linear regression → moderated regression (M5) → ANOVA (T15) → moderated mediation (T16) → RSM inverted U (B1/C1)

O — Outcome

Primary Outcome Variables

Variable	Measurement	Key Result
Q3_Diff	Post-pre gap in digital adoption	Inverted-U relationship (p = .029)
DT Infrastructure Improvement	Difference on 5-point Likert scale	Moderation effect tested

Key Findings

Finding	Statistic	Interpretation
Firm-size inverted U	p = .029	Statistically significant nonlinear relationship
Optimal size	≈ 13 employees	Balance point between resources and inertia
Large-firm paradox	Negative slope	Training effectiveness declines as size grows
SF + multiple participation	Negative synergy	Conflicting rather than simply additive effects

Analysis Matrix (Four-Method Stepwise Deepening)

Method	Analysis Target	Identification Strategy	Result
M5 Moderated Regression	Firm size × DT awareness → Q3	First-order interaction test	Moderation effect 
T15 ANOVA	Mean differences across size groups	Between-group comparison	Suggests nonlinear pattern
T16 Moderated Mediation	Moderation of mediating pathway	Mechanism identification	Differentiated mediating pathways
B1/C1 RSM	Response-surface inverted U	Nonlinear optimum estimation	Optimum ≈ 13 employees

Hypotheses

- **H1:** Firm size and DT training effectiveness have an inverted U-shaped relationship
- **H2:** An optimal size point exists, at approximately 13 regular employees
- **H3:** Large firms experience reduced training effectiveness due to organizational inertia (large-firm paradox)
- **H4:** Combining the SF track with multiple participation produces **negative synergy** rather than positive synergy

Theoretical Framework

Theory	Application
Liability of Smallness (Aldrich & Auster, 1986)	Resource and capability shortages of small firms → nonlinear extension
Resource Orchestration (Sirmon et al., 2011)	Efficiency of resource mobilization varies with firm size
Optimal Distinctiveness (Zhao et al., 2017)	An optimal position that is neither too small nor too large

Strengths / Weaknesses

Strengths

- **Clear policy implications:** Enables design of size-specific tailored support
- **Practically applicable:** Provides a concrete indicator for the optimal size (≈13 employees)
- **High novelty:** The inverted-U finding challenges the existing linear assumption
- **Highest practical value** (among all strategies)

Weaknesses and Remedies

- **Low R^2** → Add control variables + emphasize shape analysis in RSM
- **Crude firm-size proxy:** Uses only number of regular employees → consider supplementary indicators such as sales, assets, and job complexity
- **Robustness of the inverted-U finding:** Comparative testing against alternative functional forms (logarithmic, S-shaped) recommended

Key Contributions

Theoretical Contributions

- **Nonlinear extension** of Liability of Smallness (from a simple monotonic relationship to an inverted U)
- **Empirical evidence of an Optimal Level** in training effectiveness
- Verification of size-dependent efficiency differences in Resource Orchestration

Methodological Contributions

- Analytical deepening strategy from **OLS moderation to RSM inverted U**
- Four-method stepwise deepening protocol (linear → nonlinear)

- A standard analytical framework for firm size–training interactions

Policy Contributions

- Evidence for a **size-differentiated support strategy**
 - Different intervention designs for firms before and after reaching the optimal size
 - SF + multiple participation negative synergy → guidelines on overlapping track applications
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Related Files

- `~/Projects/SME-Transformation/` — project root
- Data: 282 cases × 166 columns, 13 tracks
- Core analyses: M5, T15, T16, B1/C1